Phase1

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Real Healthcare Context

Heart failure is one of the most common, costly, and life-threatening cardiovascular conditions worldwide. According to **Robbins Basic Pathology** (10th Edition), heart failure affects more than 5 million individuals in the United States alone, leads to over 1 million hospitalizations annually, and results in a financial burden exceeding $32 billion. Globally, cardiovascular diseases remain the leading cause of death, accounting for approximately one in every four deaths.

Beyond developed countries, the burden is even more critical in developing nations such as **Egypt**, where cardiovascular diseases represent a major public health challenge. In Egypt, heart disease is one of the leading causes of mortality due to factors such as:

- High prevalence of hypertension and diabetes
- Increasing rates of obesity
- Smoking habits, especially among adults
- Limited access to specialized cardiac care in rural and underserved areas

Telemedicine in Real Healthcare :

In real-world healthcare environments, especially in Egypt and similar regions, patients with heart failure symptoms often do not have immediate access to cardiologists. Instead, they rely on:

- Primary healthcare units
- Rural clinics
- Emergency hotlines
- Telemedicine and mobile health applications Patients typically report symptoms such as:
- Shortness of breath
- Chest pain
- Swelling (edema)
- Fatigue or reduced activity tolerance

These symptoms may indicate anything from **mild chronic conditions** to **life-threatening emergencies**, making early triage essential.

The proposed Knowledge-Based System is designed specifically for T**elemedicine-based Triage**. It acts as an intelligent assistant that **supports early clinical decision-making** before a specialist becomes available.

The system processes three categories of patient data:

1-Clinical Symptoms

- Shortness of breath
- Orthopnea
- Edema
- Chest pain
- Cough

- Low activity
- Palpitations
- Dizziness
- Syncope
- Chest tightness

2. Basic Health Indicators

- Blood pressure
- Heart rate per minute
- Respiratory Rate
- Oxygen saturation
- Temperature
- Hemoglobin

3. Background Information

- Hypertension (High blood pressure)
- Diabetes
- Heart disease
- Obesity
- Previous heart attack
- Chronic lung disease
- Kidney disease
- Age

System Functionality

Based on the input data, the system:

- Identifies the **Possible Conditions** :

1. Acute Decompensated Heart Failure
2. Chronic Heart Failure (Stable)
3. Heart Failure with Preserved EF (HFpEF)
4. Acute Myocardial Infarction (STEMI / NSTEMI)
5. Acute Coronary Syndrome (ACS)
6. Stable Angina
7. Hypertensive Emergency
8. Atrial Fibrillation
9. Sudden Cardiac Arrest
10. Cardiogenic Shock

-Assigns an **Urgency Level**:

- **Critical**
- **High**
- **Moderate**
- **Low**

Suggested Actions:
- Rest and avoid strenuous activity
- Rest and reduce stress
- Schedule a routine follow-up with your doctor
- See your doctor within the next few days
- Go to the emergency room immediately

Realistic Scenarios

Scenario 1: Mild Case

Patient Data:

Age: 40-60

Symptoms: Shortness of

breath (mild) **Vital Signs:**

Blood Pressure: 145/90 mmHg

Heart Rate: 80 bpm

Oxygen Saturation: 96%

Background:

Excess weight

Expected Output:

Disease : Chronic Hypertensive

Heart Disease **Urgency Level :**

LOW

Action :

Reduce stress Routine

doctor follow-up

Scenario 2: Moderate Case

Patient Data:

Age: 60

Symptoms:

Shortness of breath

Edema

Low activity ability

Vital Signs:

Blood Pressure: 150/95 mmHg
Heart Rate: 85 bpm

Background:

Hypertension

Heart disease

Expected Output:

Disease : Chronic Heart

Failure (Stable) **Urgency**

Level:

MODERATE

Action:

Rest Visit doctor within a few

days

Scenario 3: High Case

Patient Data:

Age: 65

Symptoms:

Shortness of breath

Low activity ability

Vital Signs: Heart

Rate: 120 bpm

Background:

Heart disease

Hypertension

Expected Output:

Disease : Atrial

Fibrillation

Urgency Level:

HIGH

Action:

See doctor today

Scenario 4: Critical Case

Patient Data:

Age: 50

Symptoms:

Severe chest pain

Shortness of breath

Vital Signs:

Blood Pressure: 150 mmHg

Heart Rate: 110 bpm

Background:

Smoking

Family history

Expected Output:

Disease : Acute Myocardial

Infarction **Urgency Level:**

CRITICAL

Action:

Emergency room immediately

Scope

-The system is designed to help in the early assessment of **heart-related** conditions.
-It analyzes patient symptoms, vital signs, and background information.
-The system provides possible cardiac conditions, urgency level, and suggested actions.
-It supports decision-making and guides users on whether medical attention is needed.
-The system focuses **only** on predefined heart conditions included in the knowledge base.

Limitations

-The system does not replace professional medical diagnosis.
-It can only identify conditions already stored in the knowledge base.
-The system cannot detect new or unknown diseases.
-Performance of the expert system is largely determined by the quantity and quality of knowledge in its knowledge base.
-The system relies on correct user input.
-It considers only limited symptoms and measurements.